

CSE-350 Quiz-03

ADC & DAC

Set-A

Total – 20 marks

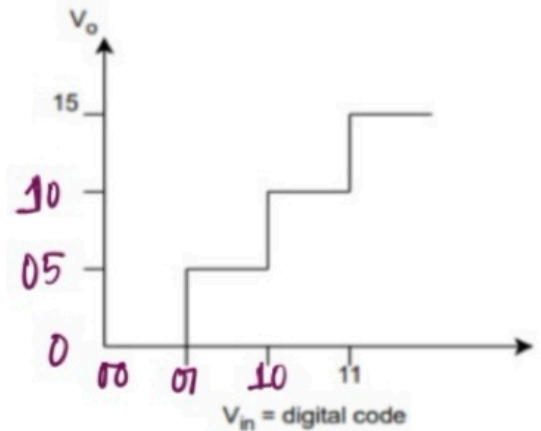
Time – 20 minutes

Name -	Id -	Section -
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Problem-1

10 marks

The figure shows the input-output characteristics of a converter. The output of the converter is denoted as V_o and the input as V_{in} . Identify the converter type and the number of bits used by the converter. Further, calculate V_{ref} when $R_F = 2R$.



Solⁿ: $\rightarrow$ DAC converter
2 bits

$$R_F = 2R$$

$$\begin{matrix} b_1 b_2 \\ 1 \ 1 \end{matrix} \rightarrow V_o = 15$$

$$V_o = V_{ref} \times \frac{R_F}{R} \left(b_1 + \frac{b_2}{2} \right)$$

$$\Rightarrow 15 = V_{ref} \times \frac{2R}{R} \left(1 + \frac{1}{2} \right)$$

$$\Rightarrow 15 = V_{ref} \times 2 \times \frac{3}{2}$$

$$\therefore \boxed{V_{ref} = 5V} \quad (\text{Ans.})$$

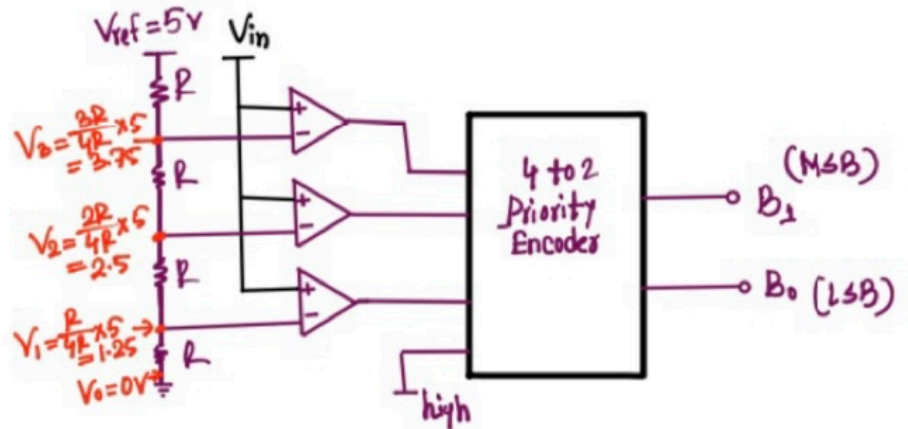
Problem-2**10 marks**

Design a 2 bit Flash ADC for an analog signal of range (0, 5). According to your circuit answer the following questions.

- What is the resolution and quantization error for this quantizer.
- Write the encoded sequences for the following input sequences { 2.3, 0.6, 4.8, 3.7 }

Soln: $n=2 \rightarrow \Delta = \frac{V_{\max} - V_{\min}}{2^n} = \frac{5-0}{2^2} = \frac{5}{4} = 1.25$
 quantization error = $\frac{\Delta}{2} = 0.625$

Quantization Range	Encoded Seq.
3.75 - 5	11
2.5 - 3.75	10
1.25 - 2.5	01
0 - 1.25	00

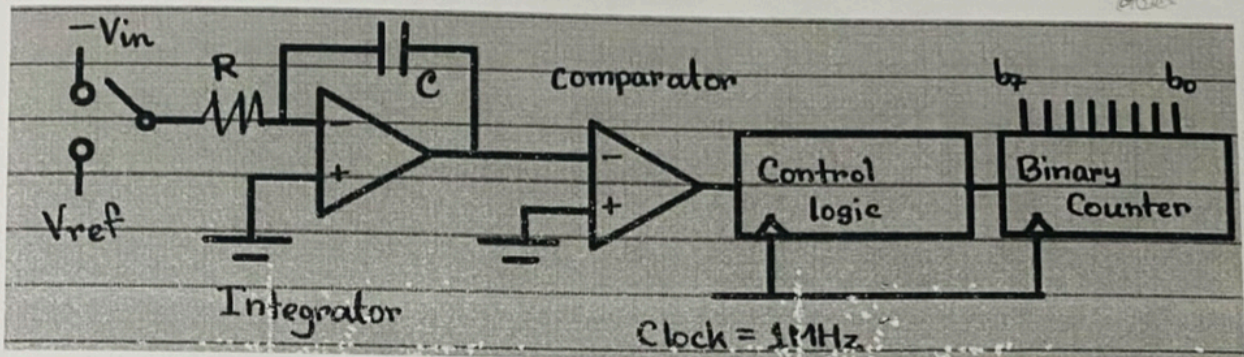


i/p $\rightarrow \{ 2.3, 0.6, 4.8, 3.7 \}$
 $\downarrow \quad \downarrow \quad \downarrow \quad \downarrow$
 encoded $\rightarrow \{ 01, 00, 11, 10 \}$

Problem-3

05 marks

A dual slope ADC has $V_{ref} = 5V$, and a 8 bit counter that outputs an 8 bit representation of the input signal. A 1 MHz clock is used for the clock and counter circuits. If the output is 01100100 answer the following questions.



(a)	Determine the input voltage range, number of steps, and step size of the ADC.
(b)	Determine the input voltage and total reading time or conversion time.

a) input voltage range = $(0 \sim 5) V$

number of step = $2^8 = 256$

step size = $\frac{V_{ref}}{2^N} = \frac{5}{2^8} = 0.01953$ (approx)

b) $V_{in} = \frac{n}{2^N} \times V_{ref}$

$= \frac{100}{2^8} \times 5$

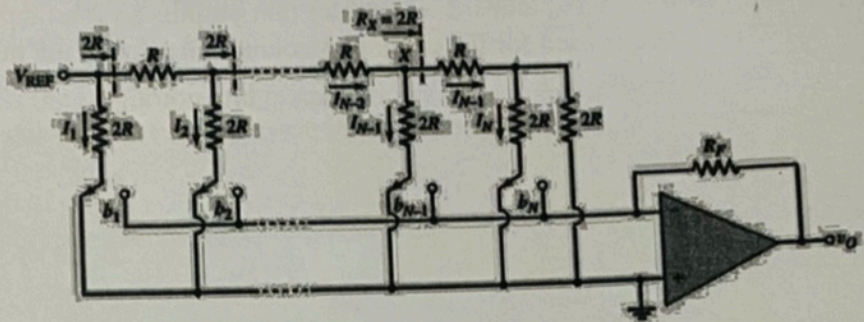
$= 1.953125 V$

$n = 100$ [decimal of 01100100]

reading time = $2^N \cdot \frac{1}{f} + n \frac{1}{f} = \frac{2^8 + 100}{1 \times 10^6}$

$= 3.56 \times 10^{-4} s$

Problem-4



The N-bit DAC with an R-2R ladder network in the above figure is to be designed as a 3-bit DAC device. Suppose $V_{REF} = -5V$ and $R_F = R = 5K\Omega$.

(a)	Determine the resolution and the currents I_1 , I_2 , and I_3 .
(b)	Draw the V_{in} Vs. V_{out} graph.

For 3 bit,

Resolution, $A = -V_{REF} \times \frac{R_F}{2R} \left(b_1 + \frac{b_2}{2} + \frac{b_3}{4} \right)$

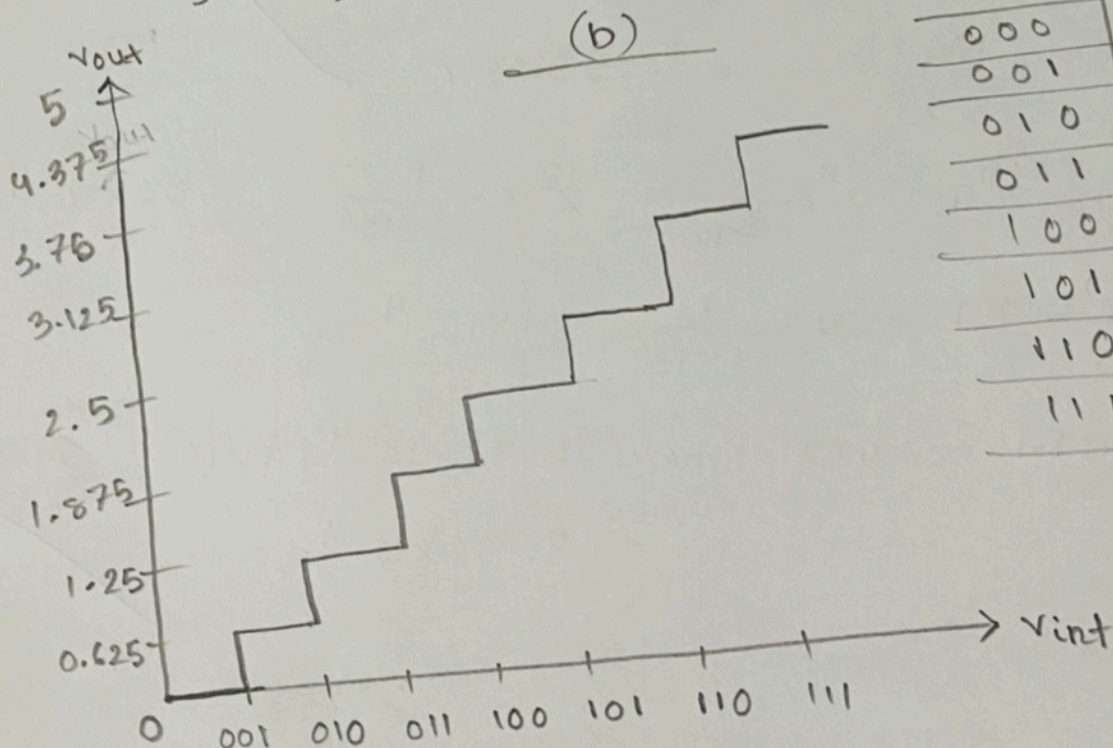
$= -(-5) \times \frac{5}{2 \times 5} \left(0 + \frac{0}{2} + \frac{1}{4} \right)$

$= 0.3125$ $\times = 0.625$

$\therefore I_1 = \frac{|V_{REF}|}{2R} = \frac{5}{2 \times 5} = 0.5 \text{ mA}$

$\therefore I_2 = \frac{I_1}{2} = \frac{0.5}{2} = 0.25 \text{ mA}$

$\therefore I_3 = \frac{I_2}{2} = \frac{0.25}{2} = 0.125 \text{ mA}$



input(v)	output(v)
000	0V
001	0.625
010	1.25
011	1.875
100	2.5
101	3.125
110	3.75
111	4.375

ADC & DAC

Set-B

Total - 20 marks

Time - 30 minutes

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Section - 6A

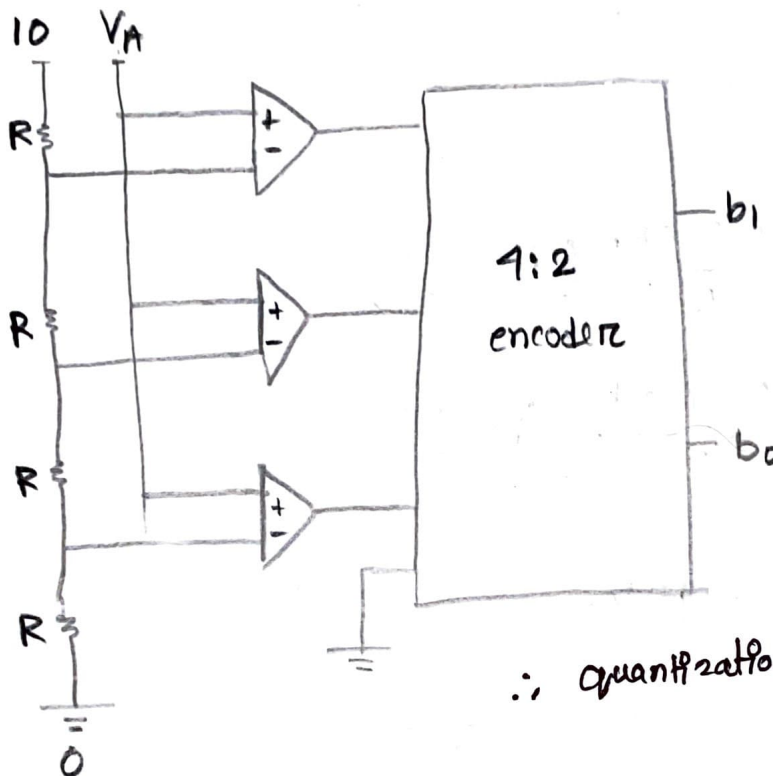
Problem-1

05 marks

Design and draw a 2 bit Flash ADC for an analog signal of range (0, 10). According to your design, answer the following questions.

- What is the resolution and quantization error for this quantizer.
- Write the encoded sequences for the following input sequences { 2.3, 7.6, 5.8, 0.7 }

(i)



$$\text{resistor} = 2^N = 2^2 = 4$$

$$\text{comparator} = 2^N - 1 = 4 - 1 = 3$$

$$\therefore \text{resolution, } A = \frac{V_{\max} - V_{\min}}{2^N}$$

$$= \frac{10 - 0}{2^2} = \frac{5}{2}$$

$$\therefore \text{quantization error} = \frac{A}{2} = \frac{5/2}{2} = \frac{5}{4}$$

(ii)

Quantization range	encoded sequence
7.5-10	11
5-7.5	10
2.5-5	01
0-2.5	00

so the given sequence will be,

$$2.3 \rightarrow 00$$

$$7.6 \rightarrow 11$$

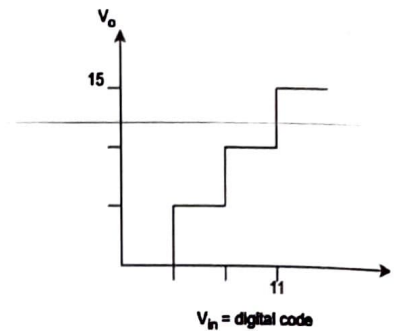
$$5.8 \rightarrow 10$$

$$0.7 \rightarrow 00$$

Problem-2

05 marks

The figure shows the input-output characteristics of a converter. The output of the converter is denoted as V_o and the input as V_{in} . Identify the converter type and the number of bits used by the converter. Further, depending on your assumption, calculate V_{ref} when $R_F = R$.



DAC
it is a BWR 1 converter.

number of bits = N

when step number $2^N = 4$

so, $N = 2$

$$V_o = \frac{V_{ref} \times R_F}{R} \left(b_1 + \frac{b_2}{2} \right)$$

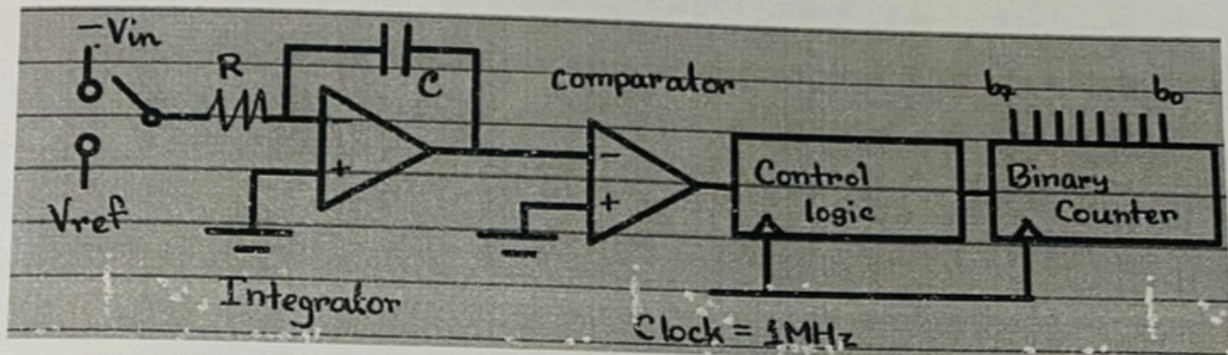
$$15 = \frac{V_{ref} \times R}{R} \left(1 + \frac{1}{2} \right)$$

$$\Rightarrow V_{ref} = 10V$$

Problem-3

05 marks

A dual slope ADC has $V_{ref} = 10V$, and a 8 bit counter that outputs an 8 bit representation of the input signal. A 1 MHz clock is used for the clock and counter circuits. If the output is 01100100 answer the following questions.



(a)	Determine the input voltage range, number of steps, and step size of the ADC.
(b)	Determine the input voltage and total reading time or conversion time.

(a)

$$N=8, V_{ref} = 10V$$

$$\therefore \text{Voltage range} = V_{max} - V_{min} = (10 - 0) = 10V$$

$$\therefore \text{number of steps} = 2^N = 2^8 = 256$$

$$\therefore \text{step size} = \frac{V_{max} - V_{min}}{2^N} = \frac{10}{256} = 0.039$$

(b)

$$\frac{V_{in}}{V_{ref}} = \frac{nT}{2^N T} \Rightarrow V_{in} = \frac{V_{ref} \cdot nT}{2^N T}$$

$$= \frac{10 \times 100 \times 1 \times 10^{-6}}{2^8 \times 10^{-6}} \quad \left| \begin{array}{l} n = (01100100)_2 \\ = 100 \\ T = \frac{1}{1 \times 10^6} \text{ s} \\ = 1 \times 10^{-6} \text{ s} \end{array} \right.$$

$$= 3.906V$$

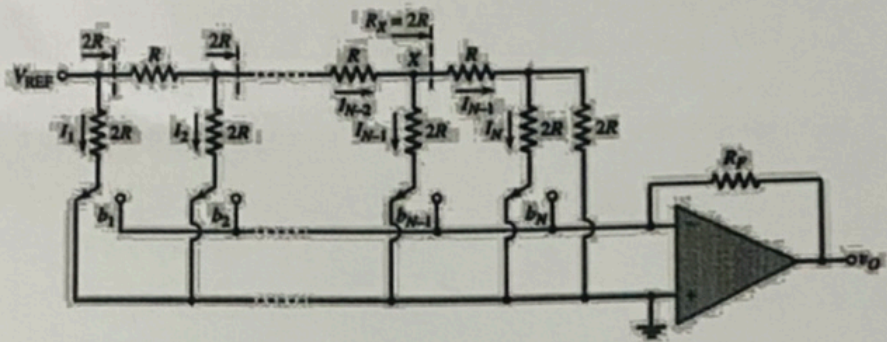
$$\therefore T_1 = 2^N T = 2^8 \times 10^{-6} = 2.56 \times 10^{-4} \text{ s}$$

$$\therefore T_2 = nT = 100 \times 10^{-6} = 1 \times 10^{-4} \text{ s}$$

$$\therefore \text{Total reading time, } T = T_1 + T_2 = 3.56 \times 10^{-4} \text{ s}$$

Problem-4

05 marks



The N-bit DAC with an R-2R ladder network in the above figure is to be designed as a 3-bit DAC device. Suppose $V_{REF} = -10V$ and $R_F = R = 10K\Omega$.

(a)	Determine the resolution and the currents I_1 , I_2 , and I_3 .
(b)	Draw the V_{in} Vs. V_{out} graph.

$$a) I_1 = \frac{V_{REF}}{2R} = \frac{-10}{2 \times 10} = -0.5 \times 10^{-3} A$$

$$I_2 = \frac{I_1}{2} = -0.25 \times 10^{-3} A$$

$$I_3 = \frac{I_2}{2} = -0.125 \times 10^{-3} A$$

b)

$$V_O = -\frac{R_F}{2R} \times V_{REF} \times \frac{1}{4}$$

$$= -\frac{10}{2 \times 10} \times -10 \times \frac{1}{4}$$

$$= 1.25$$

Step Size $\rightarrow$

